

Monalisa Dokania

Software Engineer · Distributed Systems, Cloud Infrastructure & Applied AI

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PROFILE

Software engineer with 3+ years of technical experience, focused on **distributed systems, cloud infrastructure, and applied AI**. I build fast, reliable, and thoughtfully designed production systems end-to-end.

Currently a founding engineer at an early-stage startup; alongside, I build and publish agentic AI systems.

WORK EXPERIENCE

06/2024 – present

Scottsdale, AZ, USA

Lead Software Engineer

BLUSVN Technologies (early-stage startup)

- **Founding engineer** on a distributed water-delivery platform; owned the full product lifecycle from architecture to production across **backend, infrastructure, and ML**.
- Engineered high-performance **Golang microservices** with concurrent worker pools and batched notification delivery, validated to handle **50K orders/min** under load testing.
- Architected and shipped **100+ backend APIs** (TypeScript/tRPC, PostgreSQL); reduced end-to-end latency by **40%**.
- Built real-time **ML inference pipelines** for face and ID validation, and **AI-assisted agent workflows** for log analysis that cut manual triage time on production incidents.
- Designed and operated production **AWS/EKS infrastructure** with Terraform; reduced AWS spend by **12%** and led **load and chaos testing** to validate SLOs under peak load.
- Partnered with product managers and stakeholders on technical architecture and led engineering delivery; managed CI/CD and Agile sprints.

05/2023 – 05/2024

Tempe, AZ, USA

Graduate Teaching Assistant

Arizona State University

- Supported Software Verification, Validation & Testing — **900+ students** across sections.
- Guided students in unit testing, debugging, code reviews, and OOP design principles.
- Led lab sessions and office hours; developed course materials and grading rubrics.

07/2016 – 05/2022

India

Operations & Business Analyst

M/S Rajiv Industries

- Supported day-to-day operations and data analysis at a family-run business; digitized sales records. Returned to full-time tech through Master's in Computer Science in 2022.

06/2015 – 06/2016

Noida, India

Associate Software Engineer

Indus Valley Partners Pvt. Ltd.

- Developed custom applications on the Geneva financial platform (portfolio management system used by asset managers).
- Contributed to a major platform upgrade (Geneva 8.0 → 15.2), including data migration validation and statistical reconciliation across portfolios.

TECHNICAL SKILLS

Languages	Python · Go · TypeScript · Java
Backend	FastAPI · Node.js · tRPC · Spring · PostgreSQL · Redis · Kafka
Frontend	React · Next.js · Vite · JavaScript / TypeScript
AI / ML	Machine Learning · Agentic AI · LangChain / LangGraph · RAG · Evaluation (RAGAS)
Infrastructure	AWS · EKS · Kubernetes · Docker · Terraform · Jenkins · Prometheus & Grafana
Tools	Git · GitHub Actions · JIRA · Linux

SELECTED PROJECTS

09/2025 – present	Agent-Based AI System for Document & Log Analysis Multi-agent pipeline (Planner → Retriever → Analyst → Critic → Reporter) for document and log analysis. RAG workflow over Qdrant, async processing via Redis RQ, with measurably fewer hallucinations than single-prompt approaches.  github.com/msdokania/agentic-analysis-platform
12/2025 – present	Research Assistant — Hybrid RAG + RAGAS Evaluation Phase 4 of an ongoing Agentic AI Series building LLM agent systems from first principles. Production-grade RAG with hybrid retrieval (BM25 + Semantic), cross-encoder re-ranking, and RAGAS-based evaluation against ground-truth Q&A pairs.  github.com/msdokania/agentic-ai-series

EDUCATION

08/2022 – 05/2024 <i>Tempe, AZ, USA</i>	Master of Science, Computer Science <i>Arizona State University</i> Final grade: 4.0 / 4.0
2011 – 2015 <i>Allahabad, India</i>	Bachelor of Technology, Information Technology <i>Motilal Nehru National Institute of Technology (MNNIT Allahabad)</i>

LANGUAGES

English	Full professional proficiency (C2)
Hindi	Native
German	Basic (A2)